

YOGESH KUMAR

Software Developer | Full-Stack Development | AI & Machine Learning

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PROFESSIONAL SUMMARY

BCA student and aspiring software developer with hands-on experience building full-stack web applications, educational platforms, REST APIs, and AI-powered features. Comfortable working with React, Next.js, Node.js, Python, FastAPI, MongoDB, and modern development tools.

Built practical projects involving student management, MCQ generation, exam-pattern analysis, real-time gamification, AI interview preparation, authentication, ranking systems, and microservice-based backends. Interested in solving real-world problems through scalable and user-focused software.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C

Frontend Development: React.js, React with Vite, Angular, HTML5, CSS3, Tailwind CSS

Backend Development: Node.js, REST APIs, Python FastAPI, API design, CRUD operations

AI / LLM: LLM, RAG, AI agents, automation workflows, Google Gemini API, OpenRouter

Databases: MongoDB, Mongoose, SQL, database modeling

Architecture: Microservices, API integration, authentication, caching, modular application design

Tools & Platforms: Git, GitHub, VS Code, Netlify, Vercel, Hostinger

Core Concepts: OOP, Data Structures & Algorithms, DBMS, REST architecture, problem solving

PROJECTS & DEVELOPMENT EXPERIENCE

Smart Lab & ExamPattern — Full-Stack AI & Gamified Learning Platform

Technologies: Next.js, React, TypeScript, Tailwind CSS, MongoDB, Python, FastAPI, Google Gemini API

Developed a full-stack educational platform combining exam preparation, AI interview practice, question generation, student progress tracking, and gamified learning.

- Built a Battle Arena supporting coding, mathematics, general knowledge, trivia, and prediction-based challenges with dynamic scoring, rankings, and user statistics.
- Developed Python FastAPI microservices for AI processing and battle/question-generation functionality.
- Integrated Google Gemini AI and OpenRouter for AI-powered interview preparation and educational features.
- Implemented a hybrid question-generation approach combining cached questions, database content, AI-generated questions, and algorithmic generators.
- Designed non-repeating question generation for mathematics, coding, GK, puzzles, and prediction.
- Built student profiles with achievements, skill statistics, battle history, XP, rankings, and progression.
- Integrated MongoDB using Mongoose for the Next.js application and asynchronous MongoDB access for Python services.

MCQ Generator / ExamPattern — React-Based Question Practice Platform

Technologies: React, JavaScript, Node.js, MongoDB, AI APIs

Developed an interactive MCQ practice application focused on generating questions according to selected educational topics.

- Created topic-based MCQ generation using AI generation, caching, and reusable question data.
- Designed the system to generate questions once, cache them for the active quiz session, and load them efficiently instead of repeatedly calling an AI service.
- Implemented quiz completion and result calculation with accuracy and performance tracking.
- Added logic to clear or replace previous cached questions when users select a different topic.
- Focused on reducing unnecessary AI API usage while keeping the question experience dynamic.

Student Management System

Technologies: Python, Web Technologies, REST API concepts, SQL

- Developed a web-based application for organizing and maintaining student records.
- Implemented CRUD functionality and managed roll number, name, age, email, subjects, and academic year.
- Added tracking for active, passed-out, and discontinued students.

Student Grade Application

Technologies: Python

- Created a Python application to calculate total marks, average, percentage, and grades with automated result display.

Number Guessing Game

Technologies: Python

- Created interactive number-guessing applications using random number generation, arrays, conditional logic, and limited attempts.

EDUCATION

Bachelor of Computer Applications (BCA)

Bundelkhand University, Jhansi — Currently pursuing

Class XII — Shree Saguna Inter College, Kurara, Hamirpur — 2021 — 71.5%

Class X — H.S. Convent Inter College, Kurara, Hamirpur — 2019 — 70%

CERTIFICATION

Great Learning — OOP Concepts — July 2025

SOFT SKILLS

Problem solving & analytical thinking • Quick learner • Adaptability • Team collaboration • Communication • Time management • Debugging & troubleshooting

LANGUAGES

Hindi: Fluent English: Conversational